



Core Database Concepts

Keys, Queries, Forms, and Reports

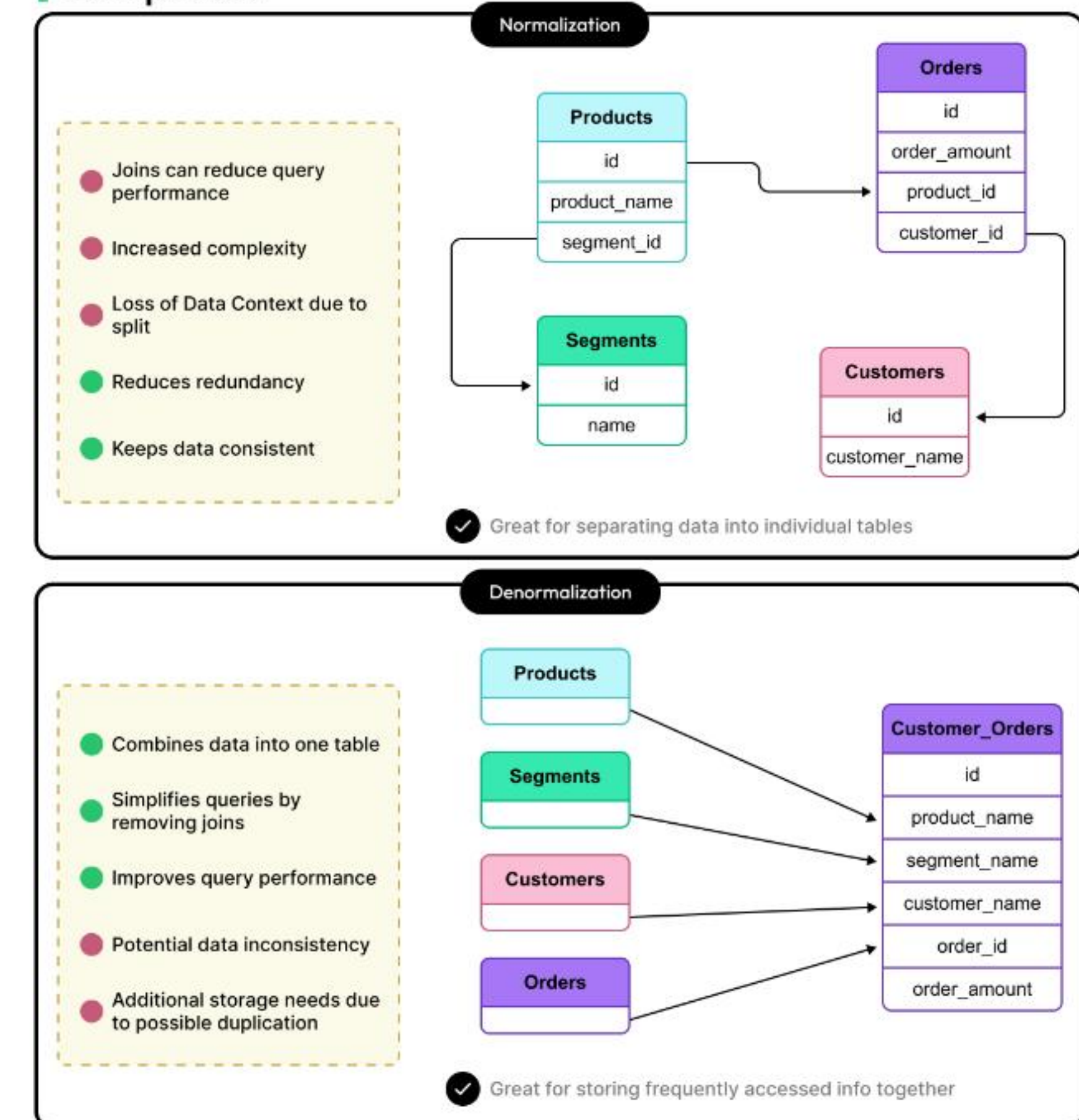
Access

What is a Primary Key?

- **Definition:** A field (or group of fields) that uniquely identifies each row or record in an MS Access table.
- **Rule 1 (No Duplicates):** Two rows cannot have the exact same primary key value.
- **Rule 2 (No Nulls):** You cannot leave a primary key field blank.
- **MS Access Behavior:** When you build a new table, Access creates an ID field set to AutoNumber by default.
- **Example:** In a STUDENT table, `Roll\_No` acts as the Primary Key.

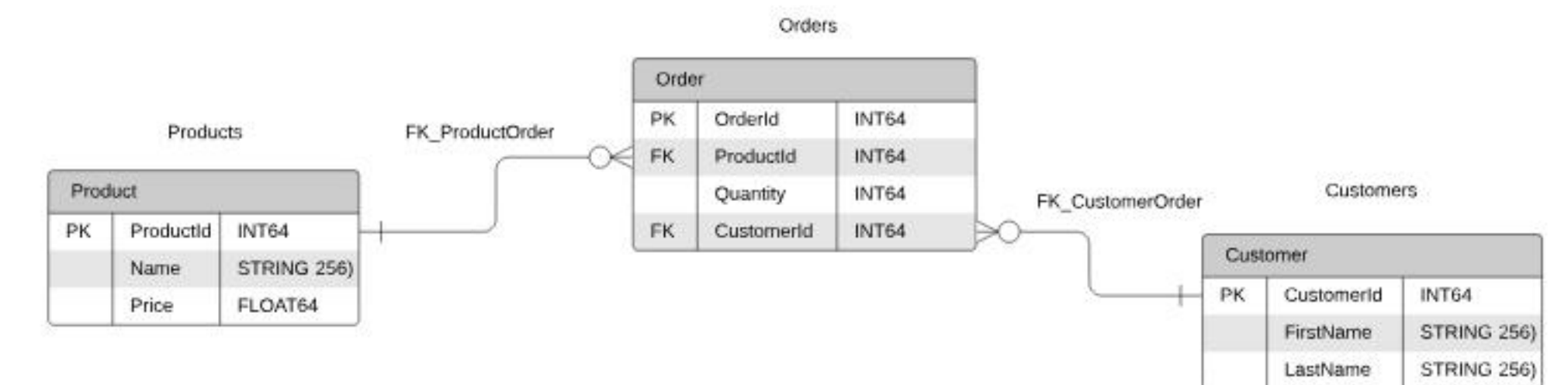
Database Schema Design Simplified

ByteByteGo



What is a Foreign Key?

- **Definition:** A field in one table that refers directly to the Primary Key of another table.
- **Rule 1 (Creates Links):** It is explicitly used to build a robust relationship between two distinct tables.
- **Rule 2 (Duplicates Allowed):** Multiple rows can hold the same foreign key value because many records might point to the same parent item.
- **Example:** A `Customer\_ID` field in an ORDERS table links back to the main CUSTOMERS table.



Primary Key vs. Foreign Key

Primary Key

- Uniquely identifies a record in its own table.
- Cannot accept duplicate values.
- Cannot contain NULL (blank) values.
- Only one Primary Key is allowed per table.

Foreign Key

- Refers to the Primary Key in another related table.
- Can easily accept duplicate values.
- Can contain NULL values.
- A single table can have multiple Foreign Keys.

Understanding Queries

A **query** is a formal request for specific data results from your database.

It allows you to ask targeted questions and pull information from one or more related tables based on specific conditions.

Crucially, queries only contain the **instructions** for fetching data; they do not store the data themselves.

Working with Forms

- **Primary Function:** Forms are used primarily for entering, modifying, and conveniently viewing records.
- **Guided Input:** The reason forms are used so often is that they are an easy way to guide people toward entering data correctly.
- **Seamless Integration:** When you enter information into a form in Access, the data goes exactly where the database designer wants it to go in related tables.

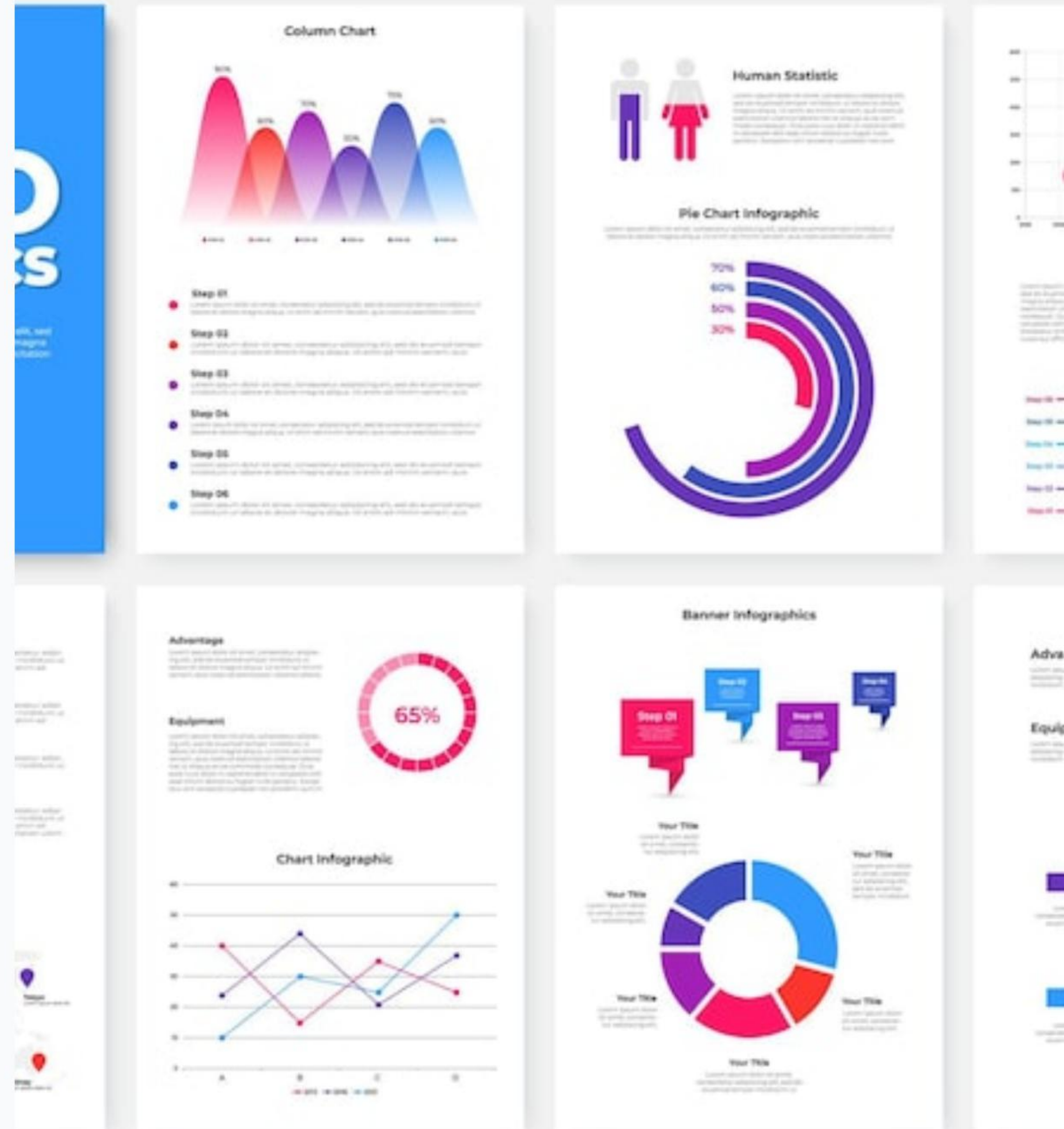
 Digital data entry form on screen

Database Reports

Reports are highly useful because they allow you to present specific components of your database in an easy-to-read format.

Access offers the flexible ability to create a report directly from any table or query.

You can extensively customize a report's appearance to make it visually appealing and professional.



Common Data Types (Part 1)

Type of Data	Description
Short Text	Text or combinations of text and numbers, including numbers that do not require calculation.
Long Text	Lengthy text or complex combinations of text and numbers.
Number	Numeric data specifically used in mathematical calculations.
Date/Time	Date and time values covering the years 100 through 9999.
Currency	Currency values and numeric data used in calculations with one to four decimal places.
AutoNumber	A unique sequential or random number automatically assigned when a new record is added.

Common Data Types (Part 2)

Data Types	Description
Attachment	Files, such as digital photos. Multiple files can be attached per record.
OLE objects	Can flexibly store pictures, audio, video, or other BLOBs (Binary Large Objects).
Hyperlink	Text or combinations of text and numbers stored as text and used as a hyperlink address.
Lookup Wizard	Creates a field that uses the contents of another table or value list to validate data entry.
Calculated	Allows you to create an expression that uses and calculates data from one or more existing fields.

Questions?

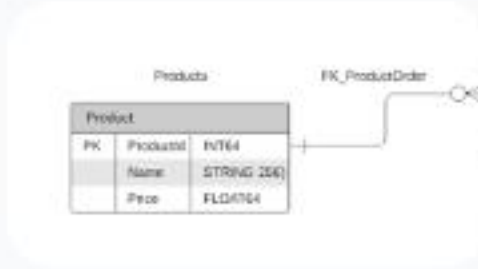
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